Network Traffic Classification: BENIGN vs DDoS

Final project report

1. Objective

Develop and evaluate machine-learning models that classify network flows as BENIGN or DDoS using flow-level traffic predictors.

2. Dataset and cleaning

The supplied dataset contains 225,745 rows and 79 columns. After removing 2,633 exact duplicate rows, 223,112 rows remained. The target classes were DDoS (128,016) and BENIGN (95,096). Numeric infinite values were converted to missing values; constant numeric features were removed; median imputation was performed inside the modeling pipeline.

3. EDA

The dataset was examined for class balance, missing/infinite values, constant features, feature distributions, class separation, correlations, mutual information, packet/timing behavior, TCP flags, and outliers. The supplied description identifies flow duration, forward/backward packet and length statistics, throughput, IAT statistics, flags, subflows, TCP window information, and active/idle statistics as key feature groups.

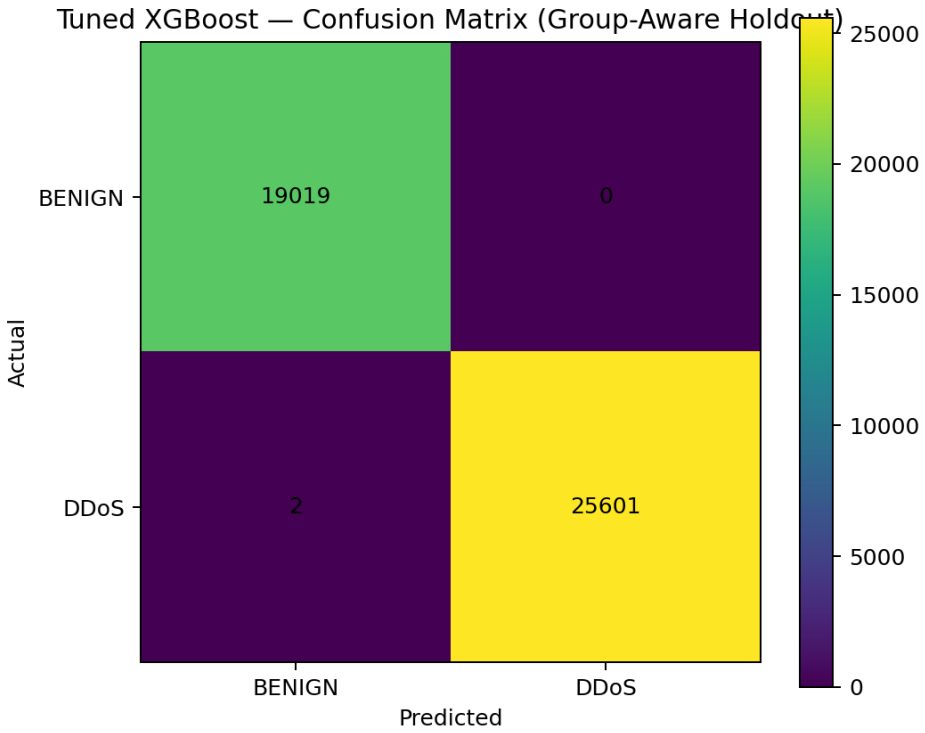
4. Modeling methodology

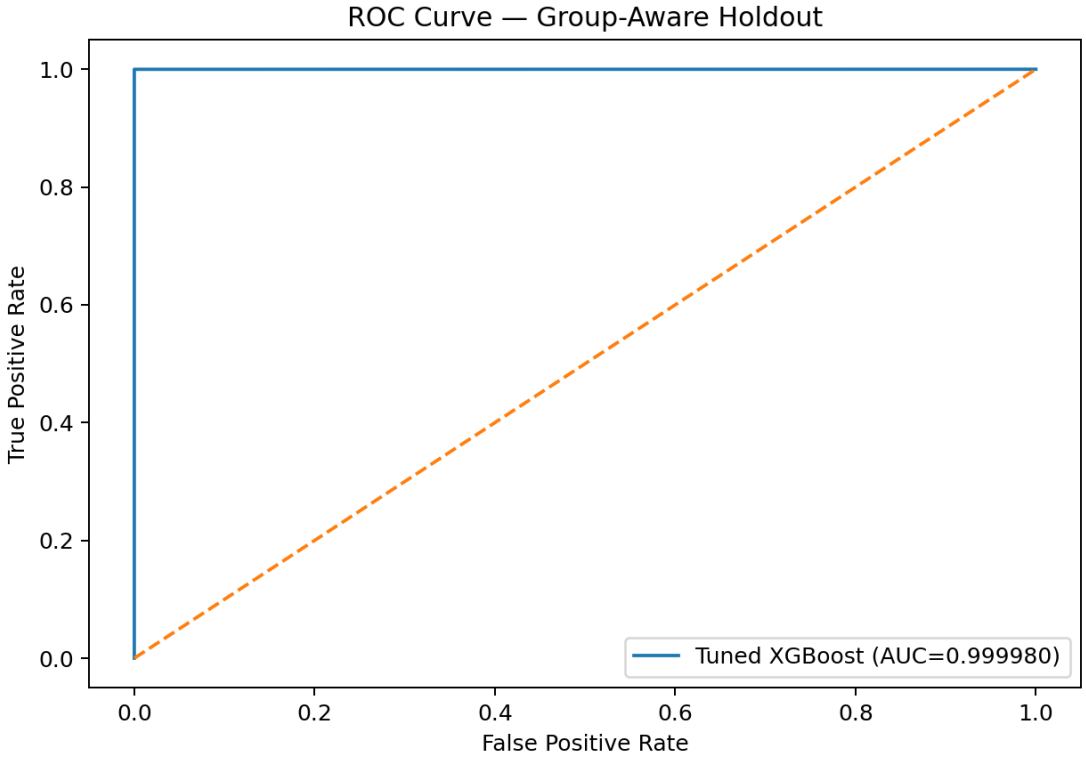
The primary evaluation uses a stratified group-aware split. Groups are identical cleaned numeric feature vectors, preventing exact feature-vector duplicates from crossing train/test. Mutual-information feature selection is fitted only on the training portion. Four base models were evaluated: Decision Tree, Random Forest, Extra Trees and XGBoost. A soft-voting ensemble was also evaluated. XGBoost was tuned using randomized search with three-fold StratifiedGroupKFold cross-validation.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Decision Tree	99.9933%	99.9961%	99.9922%	99.9941%	99.9935%
Random Forest	99.9933%	99.9961%	99.9922%	99.9941%	99.9961%
Extra Trees	99.9933%	99.9961%	99.9922%	99.9941%	99.9961%
XGBoost	99.9955%	100.0000%	99.9922%	99.9961%	99.9985%
Soft Voting Ensemble	99.9955%	100.0000%	99.9922%	99.9961%	99.9984%
Tuned XGBoost	99.9955%	100.0000%	99.9922%	99.9961%	99.9980%

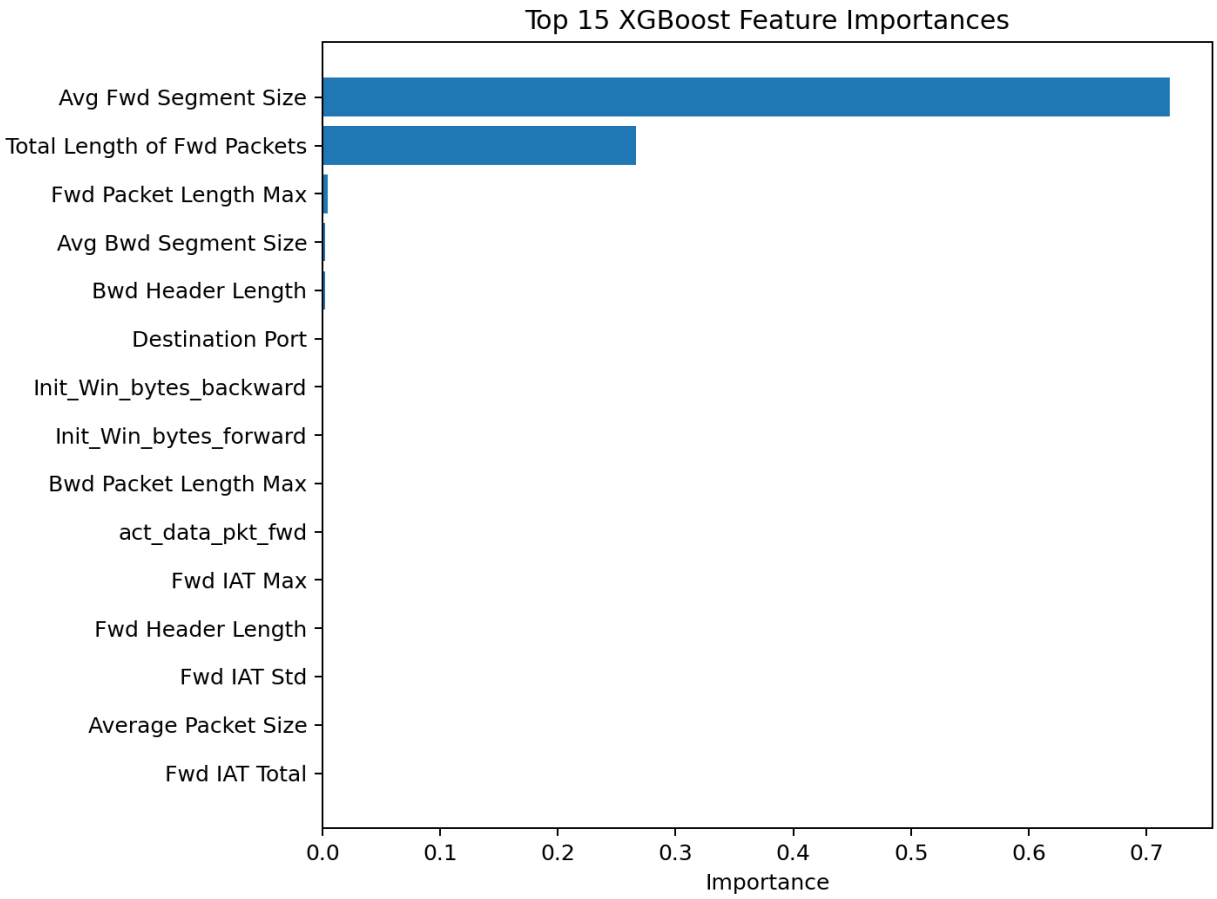
5. Final evaluation

Tuned XGBoost on the group-aware holdout achieved 99.9955% accuracy, 100.0000% precision, 99.9922% recall, 99.9961% F1 and 99.9980% ROC-AUC. The confusion matrix was TN=19,019, FP=0, FN=2, TP=25,601. The two false negatives represent DDoS flows missed by the classifier in this holdout.


Tuned XGBoost — Confusion Matrix (Group-Aware Holdout)
BENIGN
DDoS
Actual
19019
0
2
25601
BENIGN
DDoS
Predicted
25000
20000
15000
10000
5000
0


ROC Curve — Group-Aware Holdout
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Tuned XGBoost (AUC=0.999980)

6. Feature importance



7. Limitations and next steps

The supplied dataset does not expose an explicit timestamp/session/flow-ID field in the provided description, so a true time-based or capture-session generalization test cannot be established from these materials alone. The group-aware split addresses exact feature-vector overlap but does not prove performance on a completely different network. Before production deployment, validate on a separate contemporary traffic capture, monitor drift, calibrate alert thresholds, and test adversarial/previously unseen attack patterns.